Demonstration of Integrated Reservoir Operations and Extreme Hydroclimate Modeling of Water Temperatures for Fish Sustainability below Shasta Lake

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Abstract: Construction of Shasta Dam on the Sacramento River in 1945 prevented anadromous fish from accessing their natural cold-water spawning habitat above the dam and is one of the factors that caused population declines in Chinook salmon and other species. To address listing of winter-run Chinook salmon under the Endangered Species Act, the U.S. Bureau of Reclamation began operating a temperature-control device (TCD) on Shasta Dam in 1997 that enabled selective withdrawal of reservoir outflows for control of downstream water temperatures while maximizing power generation. However, effectiveness of TCD operations for addressing fisheries concerns under hydroclimate variability is unknown. Stochastic methods to generate synthetic extreme hydroclimate conditions were combined with hydrodynamic modeling of reservoir operations and interactions with reservoir managers to examine the TCD's ability to meet downstream temperature objectives. Model simulations suggest that TCD withdrawal schedules could reduce outflow temperature target exceedances when compared with more traditional dam operations such as solely hypolimnetic or epilimnetic releases. However, constraints imposed by multipurpose operational considerations, especially under extreme climate conditions, and the Sacramento–San Joaquin system's complex regulatory environment limit use of reservoir management to provide optimal thermal conditions best for Chinook salmon spawning and recruitment.
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Author keywords: Reservoir management; Sacramento River; Chinook salmon; *CE-QUAL-W2*; Temperature-control device.

Introduction

The Chinook salmon (*Oncorhynchus tshawytscha*), also known as king salmon, were once the most abundant salmon species in the Central Valley Region of California (Yoshiyama et al. 1998). They are an important commercial fishing species, especially for the Sacramento River where the first commercial salmon hatchery in the United States was established. Winter-run Chinook historically spawned during summer in cold waters of Sacramento River tributaries supplied by melting glaciers of Mount Shasta (McGinnis 1984). However, after construction of numerous dams in the Sacramento River watershed, including Shasta Dam, salmon populations dramatically declined. In 1989, the winter-run Chinook salmon were listed as a protected species under federal and California State Endangered Species Acts. The list status was changed to endangered in 1994 (Saito et al. 2001).

Shasta Dam is now operated to artificially create a cold water habitat downstream of the dam to promote salmon spawning where it did not naturally occur before dam construction (Yates et al. 2008). Reservoir operations have been altered to enable release of cold water in summer and fall to reduce salmon mortality and improve egg survival in the river (Hanna et al. 1999). The critical period for spawning and egg survival for winter-run Chinook salmon is September and October, during which being over the temperature target by only 2°C could be highly detrimental for spawning and egg survival (Kilgour et al. 1985). Reservoir managers must balance providing cold water for Chinook salmon during this critical period with the competing need to maintain the cold pool within the reservoir through late fall to manage variability in interannual cold pool replenishment (Nickel et al. 2004). Because of this, biological criteria for the health of the salmon have been developed and downstream temperature targets are enforced by the State of California.

To enable more effective downstream temperature control for fish propagation while maximizing hydropower generation, a temperature-control device (TCD) was installed on Shasta Dam and began operating in 1997 (Hanna et al. 1999). Before TCD installation, selective withdrawal to meet downstream temperature targets occurred through fixed upper outlets at 281 m above mean sea level (MAMSL) (922 ft) and lower outlets at 220 MAMSL (720 ft) (Bartholow et al. 2001), but bypassed the penstocks to the power plant at 249 MAMSL (818 ft), resulting in an estimated $63 million in lost power generation revenue from 1987 to 1996 inclusive (Hanna et al. 1999). Installation of the TCD allowed water from different elevations in the reservoir to be run through the penstocks to generate hydropower before release. A side intake

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was also installed that made it possible to withdraw water from a lower elevation than before TCD installation (Hanna et al. 1999). Simulations by Hanna et al. (1999) and Bartholow et al. (2001) indicated TCD operations reduced exceedances of downstream temperature targets in the fall.

Climate change may affect the effectiveness of using reservoir operations to manage downstream temperatures, particularly if there are increases in winter and springtime temperatures (Cayan et al. 2001). Temperatures are estimated to increase by 1.6–4.9°C (2.8–8.6°F) in the San Joaquin–Sacramento Basin over the next century depending on emissions levels (USBR 2011b; California Climate Change Center 2012). Spring snowpack in northern California is highly sensitive to temperature change, and has declined by ~23% since the 1930s (Stoelinga et al. 2010). Warming temperatures increase the likelihood for early spring snowmelt runoff and thereby increase the hazard for winter and springtime flooding. It can also prolong the duration of a summer drought (Stewart et al. 2004; Regonda et al. 2005).

Operations at Shasta are also part of the complex management of the State Water Project and the Central Valley Project in California. Reservoir managers are required to follow the operations criteria and plan (OCAP) for the Central Valley Project and the State Water Project (USBR 1997, 2008). The OCAP includes operational guidelines for reservoir and river management including flood control that are consistent with applicable laws and sensitive to impacts on listed species and designated critical habitat. These operational guidelines include water temperature requirements specific to the life history and sustainability of Chinook salmon on the Sacramento River. Managers are required to uphold the State Water Resources Control Board Water Rights Order 90-5 (SWRCG WRO 90-5) that requires downstream temperature control for winter-run salmon "to the extent controllable" (USBR 2008). SWRCG WRO 90-5 sets downstream control points with a chronic temperature threshold standard of 13.3°C (56°F) that results in fines if exceeded for 72 h or more and an acute temperature threshold standard of 13.6°C (56.5°C) that results in fines anytime it is exceeded.

One way to examine the ability of reservoir operations to meet downstream temperature needs of the Chinook salmon on the Sacramento River is through modeling. A model capable of simulating TCD operations on Shasta Lake is *CE-QUAL-W2* (*W*2). *W*2 is a two-dimensional (2D) hydrodynamic and water quality model distributed by the U.S. Army Corp of Engineers (USACE) (Cole and Wells 2011). For the project described here, the authors used a *W*2 model of Shasta Lake that was calibrated to measured water temperature data collected throughout the reservoir in 1995 (Hanna et al. 1999).

The primary research objective of this study was to examine the downstream temperature effects of outflow operations at Shasta Lake under extreme hydrologic and climatic conditions. A three-part process was used to address this research objective. Because using the historic record as modeling input may not be adequate for addressing climatic and hydrologic uncertainties associated with climate change, the project team first developed extreme hydroclimatic conditions for input into the Shasta *W*2 model using modified methods of Nowak et al. (2010). Nowak et al. (2010) developed a nonparametric, stochastic daily streamflow simulation method for generating a rich variety of synthetic flow sequences that capture the statistics of the historical data. Sapin et al. (2017) describe the completion of the first step in detail and demonstrate that it was capable of generating realistic stochastically generated data sets that fulfill the input requirements of a *W*2 model of Shasta Lake. The next steps involved simulations of Shasta Dam TCD operations in *W*2 and analysis of modeled outflow results in regards to meeting downstream and in-reservoir water temperature objectives.

A secondary objective was to engage managers in the development of the model and application of the results to management challenges. The project was designed to address the growing recognition of the need to improve the production of knowledge across scientists and stakeholders to link research results directly to decision making, as well as assist in generating new research questions (Bolte et al. 2007; Monticino et al. 2007; Cockerill et al. 2009; Voinov and Bousquet 2010). This is particularly important for developing adaptation options in water resource management given the uncertainty of future climatic conditions and the challenges to current management arrangements (Voinov and Gaddis 2008; Dilling and Lemos 2011; Kirchhoff et al. 2013). A number of approaches for engaging stakeholders in model development have emerged (Jonsson et al. 2007; Voinov and Bousquet 2010). In all approaches, modeling is viewed not merely as a technical endeavor, but as a means for engaging management decisions (Korfmacher 2001; Gaddis et al. 2010). A series of meetings were arranged to seek input from managers and modifications were made to the modeling approach to account for feasible choice sets under different scenarios.

Data and Methods

Site Description

Located 16 km (10 mi) north of Redding, California, on the Sacramento River (Fig. 1), Shasta Lake is the largest storage reservoir in California with a surface area of 119 km^2 (29,500 acres), a maximum depth of 157 m (515 ft), and a storage capacity of more than 5,550 million m^3 (4.5 million acre-feet) (Higgs and Vermeyen 1999). Shasta Lake stores and releases water for irrigation, flood control, navigation control, fish conservation, municipal and industrial use, and protection of the Sacramento–San Joaquin Delta from saltwater intrusion. Inflows to the reservoir are primarily from four tributaries: the Sacramento River, the McCloud River, the Pit River, and Squaw Creek. The reservoir also supports a popular fishery of both cold water and warm water species. On the far shore of the Sacramento River below Shasta Dam is the Livingston Stone Fish Hatchery, which is involved in fish culture related to winter-run Chinook salmon recovery (USBR 2011a).

Shasta Dam has a maximum outflow capacity of 5,267 m^3/s (186,000 ft^3/s). There are 18 outlets on the face of the dam, each 2.4 m (7.9 ft) in diameter, and a spillway. The power plant below the dam has five penstocks carrying water from the dam with an installed capacity of more than 500,000 kW (USBR 2012). Water is usually released through the penstocks of the hydroelectric plant, which has a combined maximum capacity of 498 m^3/s (17,587 ft^3/s) (Higgs and Vermeyen 1999).

Engagement with Decision Makers

The ongoing drought in California has placed increased emphasis on the need to take advantage of the flexibility of reservoir operations with the TCD at Shasta Lake to satisfy multiple demands for water in the reservoir (Service 2015). This project had two explicit goals: to understand cold pool dynamics and downstream water temperatures under stochastic extreme climate conditions and to apply methods to better link science to management. For this project, science was linked to management in three different formats.

First, a workshop was organized that introduced the project to organizations active in management, operations, and use of Shasta

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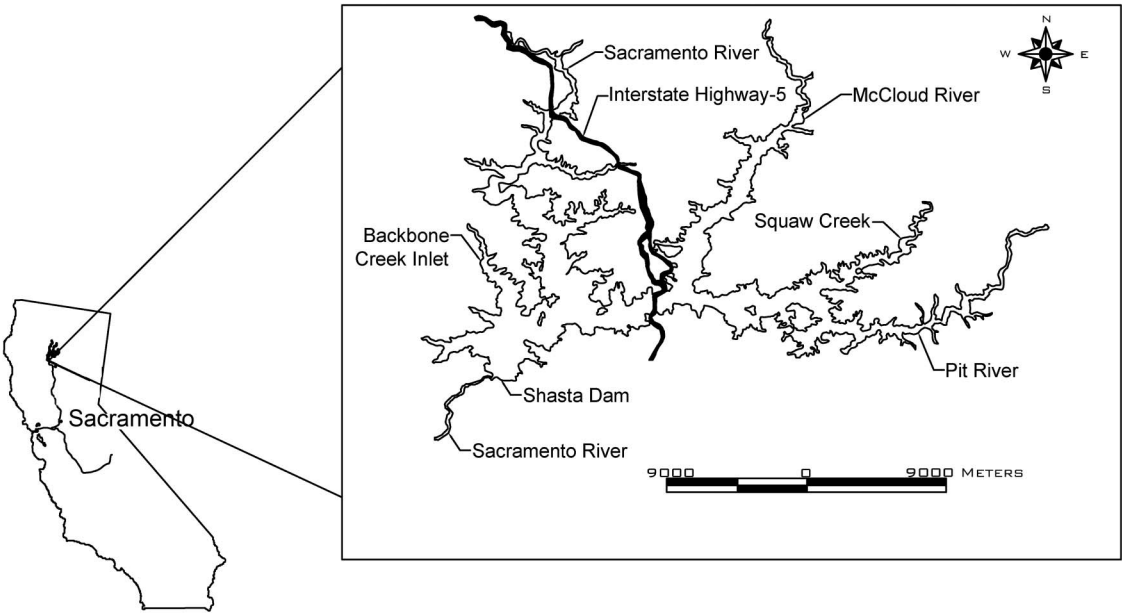

Fig. 1. Project location and Shasta Lake *CE-QUAL-W2* model extent

Reservoir. The workshop was attended by representatives in the region including managers and operators, state fish and wildlife departments, environmental advocacy groups, and tribal organizations. The project team presented a broad introduction to the project to the group. Managers of the reservoir also presented an overview of operations. The workshop ended with a structured discussion around specific aspects of reservoir operations most likely to be affected by future climate scenarios. The project team was shown how the reservoir is managed as part of the larger California system, and what information is used to determine release timing and duration. The project team was provided explanations of the regulatory framework. The procedures reservoir managers used to make operational decisions when faced with multiple priorities and how information is used in these decisions were also discussed.

A second meeting was conducted primarily with managers and operators exclusively from the U.S. Bureau of Reclamation (USBR) and National Oceanic and Atmospheric Administration (NOAA) to share preliminary modeling results and further refine the parameters of the climate scenarios generated and their interactions with operations. This additional information permitted scenarios to be adjusted to better represent the actions reservoir managers would be mandated to make in extreme climate situations.

Finally, individual interactions between the modeling team and operations managers occurred via phone calls and e-mails to refine final details. Interactions between project team members, managers, and operators, and the broader community of stakeholders allowed a closer coupling of the climate scenarios with operational rules and a fuller discussion with managers of how potential climate scenarios might influence the current management system.

CE-QUAL-W2 Calibration

The *W*2 model of Shasta Lake described in Bartholow et al. (2001) and Hanna et al. (1999) was upgraded to Version 3.7 and recalibrated to 1995 data. Resulting calibration parameters were the same as in Hanna et al. (1999) and resulted in an average R^2 of 0.966 and average root-mean-squared error (RMSE) of 0.947°C for 931 observations. The model was also verified by comparing modeled water temperatures to measured temperature profiles at the dam in 2001, 2002, and 2003 (average RMSE = 1.217°C; average R^2 = 0.842). Unless stated otherwise, all boundary conditions of the *W*2 model were the same as in Hanna et al. (1999).

Data for Model Inputs

Calendar-year simulations were run that started on January 1 (Julian Day 1) that required daily inputs of inflows, inflow temperatures, and outflows to *W*2 to assess the ability of using TCD operations to meet temperature guidelines. To capture diurnal variations in meteorology, hourly inputs of meteorology were used. The *W*2 initial conditions also required initial water temperatures and a starting day water surface elevation. Data for stochastic generation of hydroclimate sequences for *W*2 model inputs were obtained from USGS, the California Data Exchange Center, the National Oceanic and Atmospheric Administration, and the U.S. Bureau of Reclamation (Table 1) (Sapin et al. 2017).

Hydroclimate Scenario Generation

Daily streamflow, temperature, and site meteorology for the W2 model were generated using a *k*-nearest neighbor (*k*-NN) bootstrap resampling approach of Nowak et al. (2010) and Lall and Sharma (1996). Daily streamflow at all locations was generated using the method of Nowak et al. (2010), and stream temperature was generated using a *k*-NN resampling conditioned on the simulated daily streamflow (Lall and Sharma 1996). The daily meteorology was similarly resampled using *k*-NN of the simulated daily streamflow, thus ensuring that the simulated daily streamflow, stream temperature, and meteorology are all consistent. Methods for simulating these variables are described in Sapin et al. (2017).

Extreme hydrologic conditions [i.e., an extreme wet year baseline (WBL) and an extreme dry year baseline (DBL)] were selected as the year corresponding to the 99th and 1st percentiles of the aggregate annual flow, respectively. Corresponding stream temperatures and weather were then stochastically generated based on the magnitude of incoming daily streamflow. The stochastic methods produced streamflows outside the range of the historical record while maintaining the statistical nature of the system (Fig. 2). A full

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Table 1. Sources of Data for Stochastic Input Generation

Data type	Source	Period of record
Streamflow	USGS (2017): McCloud River (11368000), Pit River (11365000), Sacramento River (11342000), Squaw Creek[a] (11365500)	January 1, 1994 to September 30, 2011
Stream temperature	California Data Exchange Center (2014): McCloud River[b] (MSS), Pit River (PMN), Sacramento River (DLT)	January 11, 1989 to September 13, 2012
Air temperature, relative humidity, wind speed, wind direction, cloud cover	NOAA (2014): Redding Municipal Airport (AWS 725920)	January 1, 1994 to December 31, 2010
Computed daily inflow, reservoir storage, reservoir elevation, outlet releases	USBR[c]: operations data at Shasta Lake, California	January 1, 1994 to December 31, 2010

[a]Squaw Creek record of streamflow ends in 1966 (representing ~2% of total lake inflow); later flows are assumed to be 2.1% of total inflows based on historical proportions.
[b]McCloud River stream temperatures were used for Squaw Creek because Squaw Creek stream temperatures were not available.
[c]Computed daily inflows represent total inflows to the reservoir after adjustment for precipitation and evaporation.

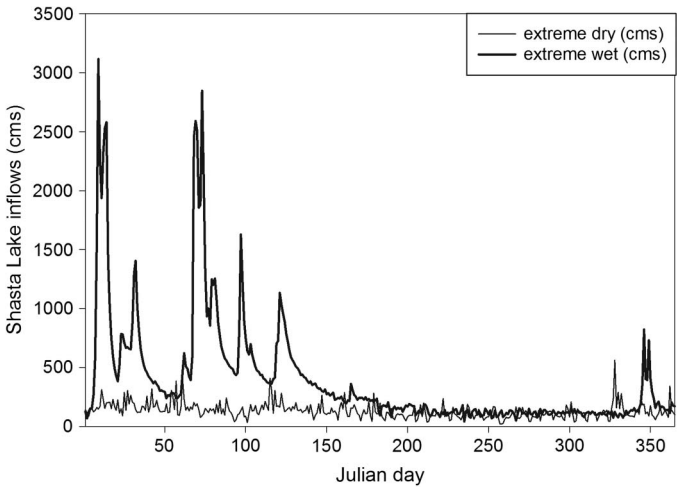

Fig. 2. Inflows to Shasta Lake for extreme dry and extreme wet years

description of the stochastic methods and data used to generate the inputs is provided in Sapin et al. (2017).

Conditions of increasing air temperatures under climate change were modeled for the stochastically generated extreme wet and extreme dry scenarios. Two ranges of projected average temperature by 2100 increase to the Sacramento–San Joaquin Delta based on emissions were presented in California Climate Change Center (2012). The project team used California Climate Change Center (2012) estimates of future air temperature for high emissions (2.6–4.9°C) and low emissions (1.6–3.3°C) to set up climate scenarios by increasing air temperatures for extreme wet [wet with high emissions climate change (WHCC) and low emissions climate change (WLCC)] or dry scenarios (DHCC and DLCC, respectively). Monthly temperature increases from California Climate Change Center (2012) were applied to air temperatures for the 15th day of each month and linearly interpolated for days in between.

The project team also collaborated with reservoir managers to develop modeling scenarios of conditions they found challenging. The managers suggested modeling scenarios of multiyear drought and unanticipated fall conditions (i.e., wet year with dry fall or wet year with warm fall). Unanticipated fall conditions are challenging because cold water storage may not be sufficient to cover the fall transition period when air temperatures may still be high. A 2-year drought scenario was developed by repeating the inputs of the extreme dry year for an additional year (D2YD). A wet year with a dry fall (WYDF) was created by combining the first 224 Julian days and Julian Days 326–330 of the stochastically generated streamflows of the extreme wet year with the remaining days of the extreme dry year (the stochastically generated extreme dry year included a large storm during Julian Days 326–330). Julian Day 224 (August 11) was chosen because streamflow for both the extreme wet and extreme dry scenarios were approximately the same on that day, and the timing was deemed appropriate by reservoir managers. Selection of Julian Day 224 for the transition to dry conditions also minimally altered the stochastically generated hydrographs to maintain the statistical nature of the system (Sapin et al. 2017). A wet year with a warm fall (WYWF) was developed by forcing the high emissions temperature increases for the extreme wet year onto the meteorology of the extreme wet year for the critical salmon rearing months of September and October.

*W*2 requires initial conditions for starting day water surface elevation that the modeling team based on *CalSim II* average storage figures for wet and dry years (CDWR 2013). The resulting Julian Day 1 water surface elevations for these storages were 280 and 309 m (919.3 and 1,013.3 ft) for the extreme dry and extreme wet year scenarios, respectively. Julian Day 1 water temperatures were set at 9°C throughout the reservoir profile as was done in Hanna et al. (1999).

Generation of Outflow Quantities

Reservoir managers provided the flood-control curve for Shasta Reservoir, noting that in wet years they are mandated to keep storage at or below the curve. The curve includes decision points, but because the extreme wet year contained inflows beyond what had been seen in the historical record, the most conservative flood-control curve that anticipates the heaviest inflows was assumed (Fig. 3). Outflow quantities were determined by calculating releases needed to maintain the flood-control curve, but subject to also satisfying minimum and maximum flow criteria. Minimum flows of 90 m^3/s (3,250 ft^3/s) must be maintained throughout the year, and the maximum allowable outflows are 2,237 m^3/s (79,000 ft^3/s) to prevent flooding downstream (USBR 2008). Furthermore, the OCAP specifies ramping criteria to prevent large changes in flow quantities in a short period of time (USBR 2008). After consultation with reservoir managers, these criteria were simplified to changing flows by no more than 15% per day.

An additional constraint that was maintained for extreme wet year scenarios was the portion of the Fall X2 requirement (USBR 2008) that requires managers to keep Shasta storage from

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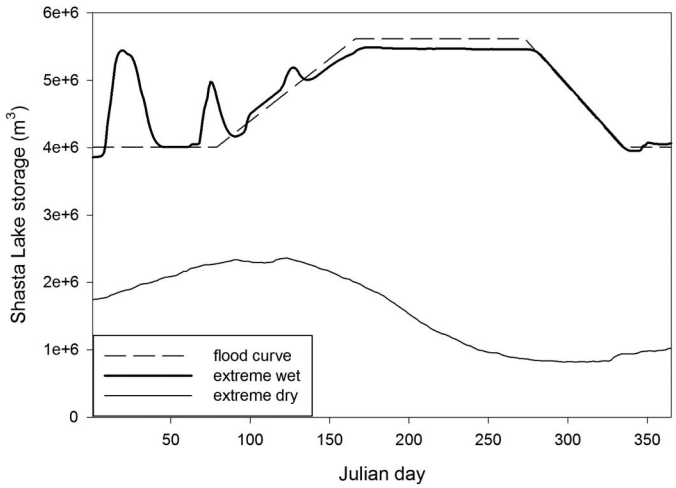

Fig. 3. Storage curves for Shasta Lake simulations

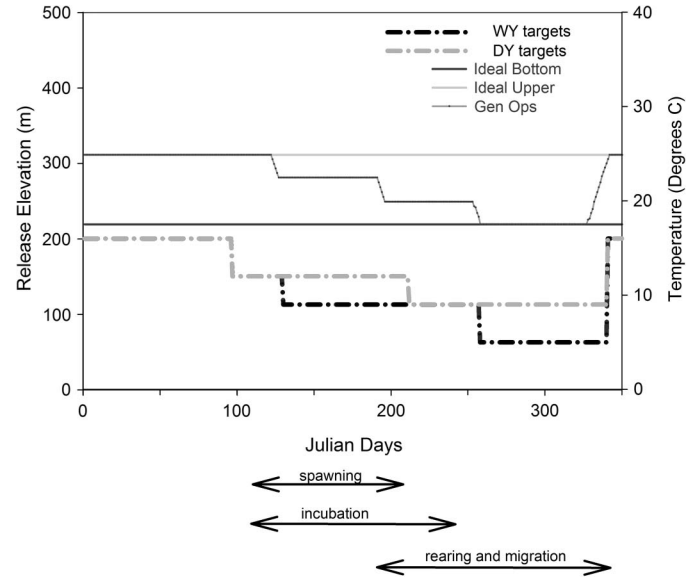

Fig. 4. Temperature targets for extreme wet (WY) and extreme dry year (DY) scenarios and simulated operations schedules over the calendar year; arrows below figure indicate time frames given in Bartholow (2004) for winter-run Chinook salmon spawning, incubation, and rearing and migration

increasing during October in a wet year to help maintain salinity conditions for Delta smelt. Therefore, any water that entered the reservoir in October was passed directly downstream.

Monthly outflow distributions were provided by reservoir managers for the extreme dry year scenario. These flows were also subject to the minimum and maximum flow requirements and the ramping criteria that were applied to the wet year flows.

Generation of Outflow Elevations

Three calendar year schedules of TCD operation were modeled for each scenario. The first was to release as much water as possible out the lowest elevation gate of the TCD (centerline elevation at 219.5 m) throughout the year to mimic a dam with hypolimnetic withdrawal; this option was referred to as the bottom operations schedule. Outflow capacity ranged between 410 and 495 m^3/s (14,500 and 17,500 ft^3/s) depending on elevation. Wet year outflows exceeding TCD capacity were directed to the bypass outlets and ranged between 345 and 2,237 m^3/s (12,200 and 79,000 ft^3/s).

The second schedule involved releasing all water out of the highest TCD gate possible without violating minimum submergence criteria of an active TCD gate [i.e., a fully open gate must be submerged at least 10.7 m (35 ft) above the bottom of gates at elevations 219.5, 249.3, and 281.3 m, and at least 6.1 m (20 ft) above the bottom of gates at elevation 311.5 m]. This schedule was referred to as the uppermost operations schedule. As with the bottom schedule, if TCD capacity was exceeded, excess flows were directed to go through the bypass outlets.

The third schedule was called a gen ops schedule that involved releasing from different TCD elevations throughout the year. In this simplified version of using the TCD, the schedule was oriented around releasing from one elevation at a time except when transitioning between gates. One generalized operations release schedule was constructed for both extreme dry and extreme wet year scenarios by considering the timing of the temperature target schedules used to evaluate model outputs (discussed in "Assessment of Model Output") and historical elevation transitions of the TCD gates for 1997–2011. Historical operations showed that gate elevation changes were usually not made abruptly, so the project team applied a 6-day linear transition schedule each time gate elevations changed. Adjustments were made to the schedule to satisfy gate submergence criteria, and bypass outlets were used when TCD capacity was exceeded.

Each of these schedules (bottom, uppermost, and gen ops) was simulated for each scenario, resulting in three operations simulations for each scenario. For the 2-year drought, the generalized operations schedule was run for the first simulated year and then the three schedules were applied to the second year.

Assessment of Model Output

An assessment of outflow temperatures was conducted to examine if TCD operations at Shasta Lake can assist with meeting federally mandated monthly downstream temperature targets for Chinook salmon on the Sacramento River in California under the scenario conditions. Actual temperature targets are not evaluated directly downstream of Shasta Dam, but rather at different points along the river depending on climate and flow conditions (USBR 2008). Reservoir managers advised using the temperature targets for the Future No Action *CalSim II* simulation that has four tiers of temperature targets based on 82 years of end-of-May Shasta storage levels (USBR 2008). It was assumed that the extreme dry year would fall within Tier I temperature targets with end-of-May Shasta storage less than 3,824 million m^3, and the extreme wet year would fall within Tier IV temperature targets with end-of-May Shasta storage greater than 5,057 million m^3 (Fig. 4).

Reservoir managers consider degree-days in managing Shasta operations, which were used as a metric for how well simulated operations of the Shasta Dam TCD can meet downstream temperature requirements in Fig. 4. Degree-days measure both the degrees above the target temperature and the number of days over the target temperature. For example, both 1° above the target temperature for 2 days and 2° above the target temperature for 1 day would result in a calculation of 2 degree-days above the target temperature. Use of degree-days also corresponds to how exceedances above required stream temperatures for Chinook salmon are understood in the Sacramento River. The sums of degree-days over the temperature target between April 7 and December 7 (Julian Days 97 and 341, respectively) were calculated.

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To analyze the ability of TCD operations to preserve in-reservoir cold water, the volume of water in the reservoir at or below the lowest temperature target [9°C (48°F)] was calculated weekly for each calendar year simulation. The volumes of the May 28 (Julian Day 148) and November 1 (Julian Day 302) cold water pool were also determined. The cold pool temperature threshold and the dates of evaluation were provided by reservoir managers.

Results

Assessment of Degree-Day Exceedances

Simulation results indicated that degree-day exceedances occurred for all scenarios regardless of reservoir operations schedule [Fig. 5(a)]. Overall, the generalized TCD operations schedule did result in slightly fewer degree-day exceedances for most scenarios that the all-out-the-uppermost or all-out-the-bottom operations schedules. Extreme wet scenarios did result in much fewer degree-day exceedances except when applying the all-out-the-uppermost releases. Because releases out of the uppermost outlets would release the warmest water throughout the year, it is not surprising that this release schedule would result in the most degree-day exceedances for all scenarios. With climate change, degree-day exceedances increased slightly for both the extreme wet year and extreme dry year [Fig. 5(a)].

Examination of daily time series of simulated outflow temperature output from *W*2 revealed more details of how reservoir operations affect outflow temperatures. Because of submergence criteria, releases were forced to move to a lower elevation when the reservoir elevation dropped too low. This is illustrated most dramatically in the extreme dry year (Fig. 6), where the sharp drop in outflow temperature around Julian Day 258 and Julian Day 274 for the generalized TCD operations and the all-out-the-uppermost TCD schedule (Fig. 6), respectively, is due to a violation in the submergence criteria that required releases to be made from a lower gate.

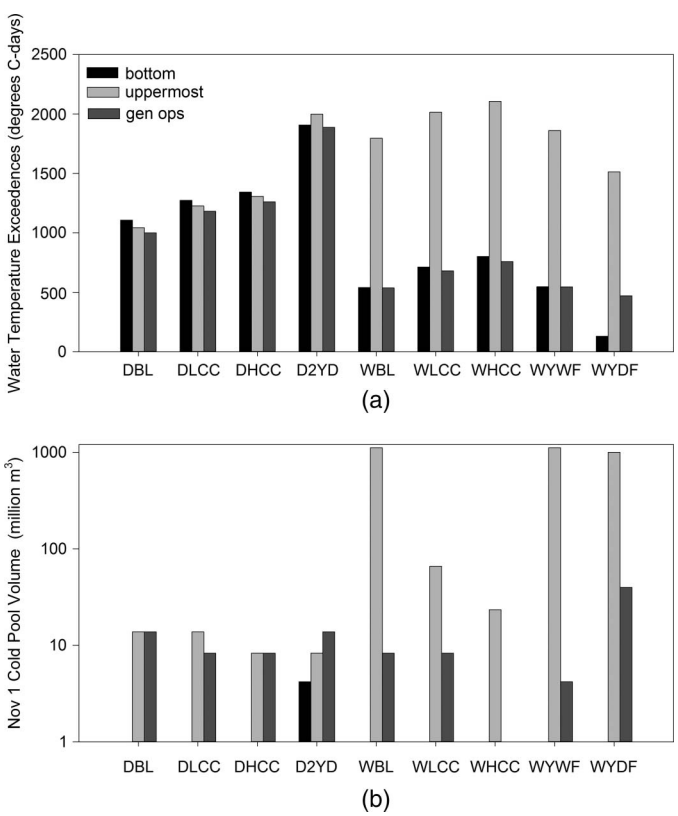

Fig. 5. Comparisons of (a) degree-day exceedances; (b) November 1 cold pool volumes (note the log scale) for all scenarios under the three operations schedules; the best option would minimize degree-day exceedances and maximize cold pool volume

Assessment of Cold Water Pool Volumes

As shown in Fig. 5(b), the November 1 cold pool volumes for all extreme dry scenarios under all operations schedules were very low, and only the releases out the uppermost gates resulted in larger November 1 cold pool volumes for the extreme wet year scenarios. Climate change scenarios resulted in less November 1 cold pool volume than their baseline counterparts. Plots of the weekly time series of in-reservoir cold water pool below the cold pool temperature threshold (9°C) for each TCD schedule for the extreme dry and extreme wet scenarios (Fig. 7) show that the in-reservoir cold water reserve is almost completely gone by Julian Day 200 (mid-July) for the all-out-the-bottom release schedule regardless of whether it is an extremely wet or dry year.

Discussion

Model results indicated the generalized operations TCD schedule most often minimized degree-day exceedances over temperature standards and was most appropriate in controlling downstream temperatures for the extreme dry year (also the 2-year drought). However, for the extreme wet year, the wet year with a dry fall, and the wet year with a warm fall, the simulated all-out-the-bottom TCD schedule minimized degree-day exceedances over the chronic and acute standards. This suggests that the in-reservoir cold water pool storage for the extreme wet year was large enough to maintain low-elevation cold water releases throughout the calendar year even during continuous depletion, whereas cold pool storage in the extreme dry year was not.

As expected, according to model simulations, the extreme dry year presented management challenges because reservoir storage was so depleted that many of the outlet elevations were unavailable for use. Thus, management of temperature by using generalized operations did not improve cold pool storage or reduce degree-day exceedances very much when compared with management strategies like all-out-the-bottom and all-out-the-uppermost release schedules. Temperature-control operations options for an extreme dry year are therefore both limited and less effective than for an extreme wet year (Fig. 5). The simulation of the 2-year drought in which general operations were used for the first year followed by the different operations schedules in the second year interestingly showed that the adjustments to operations in the second year could improve cold pool storage [Fig. 5(b)], but did not help with reducing degree-day exceedances. The inability to manage downstream temperatures and cold water storage during extreme dry years could result in significant ramifications for providing a suitable habitat for salmon migrations and reproduction. Further examination is recommended of the limitations of the effectiveness of managing elevations or reservoir releases in dry years and the threshold at which such management of reservoir operations no longer has an effect on degree-day exceedances.

Intuitively, one might expect better cold pool storage in an extreme wet year due to having larger volumes of water to work with, and hence more ability to manage downstream water temperatures. Nickel et al. (2004) suggested that reservoir management to increase reservoir volume to capture cold water inflow before

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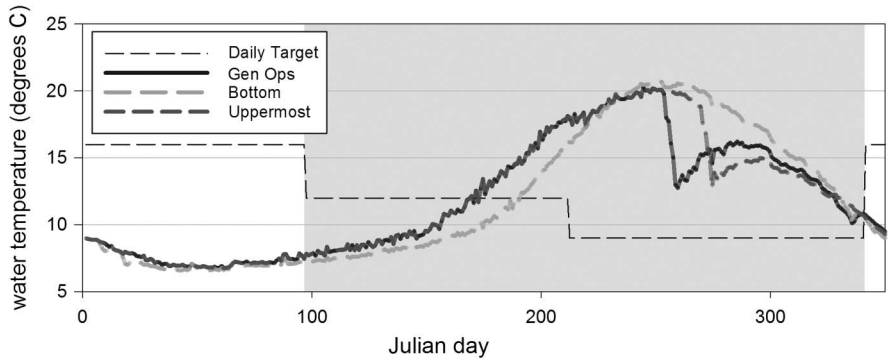

Fig. 6. *W*2 simulated outflow temperatures and the downstream temperature targets for the extreme dry year from January 1 (Julian Day 1) to December 31 (Julian Day 365); the shaded area is the time period over which degree-day exceedances were calculated; lines for the results of the different operations schedules overlap at some times of the year and may appear as one line

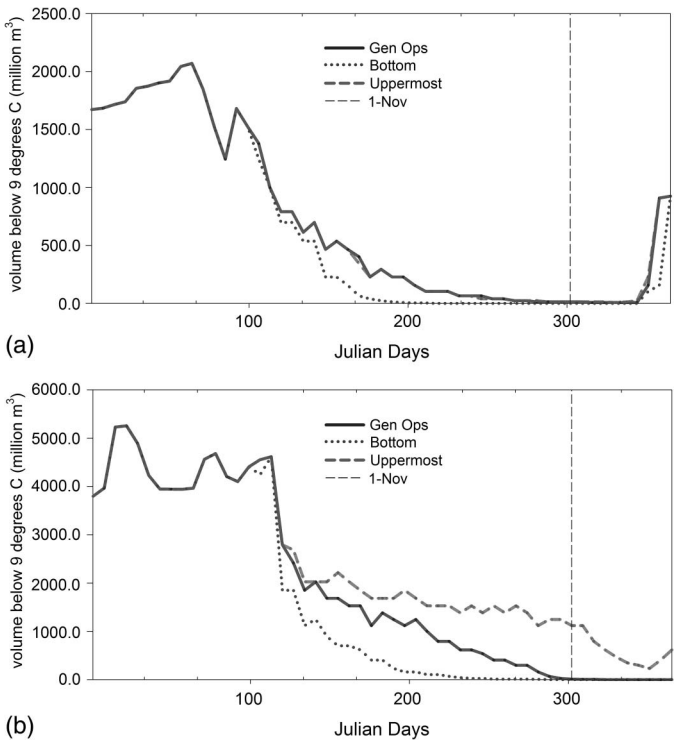

Fig. 7. Volumes of cold water pool for the (a) extreme dry year scenario; (b) extreme wet year scenario

May could improve operational flexibility at Shasta Lake. However, flood-control operations for the extreme wet year mandate that a large amount of potential storage is released during the spring. The situation is further complicated by the Fall X2 requirement that requires reservoir managers to release inflows in October to manage salinity in the delta. This requirement reduces the amount of control that operators have during a critical period of cold water management. Furthermore, limitations of penstock capacity require release of large flows through the bypass outlets, thus reducing the amount of temperature control during large releases. Despite these limitations, model simulations indicated the cold water reserve was large enough during the extreme wet year that the all-out-the-bottom TCD schedule did minimize degree-day exceedances. The better performance of the all-out-the-bottom TCD schedule as compared with the generalized operations schedule for the extreme wet year, however, was achieved only by the trade-off of a massive amount of November 1 cold water pool depletion when compared with the generalized operations and all-out-the-uppermost TCD schedules (Fig. 5). The simulated all-out-the bottom operations almost completely depleted the cold water pool by mid-July [Fig. 7(b)] so that cold water temperatures for winter-run Chinook salmon incubation, rearing, and migration may not be available under this scenario (Bartholow 2004). This trade-off highlights the importance of analyzing multiple metrics when evaluating the performance of TCD operations.

The degree to which seasonal variations might influence downstream conditions can be difficult to predict and are complicated by competing management requirements. Model results indicated that seasonal variations for the extreme wet year (i.e., WYWF and WYDF) did slightly reduce degree-day exceedances [Fig. 5(a)], but had mixed results on cold pool volumes [Fig. 5(b)]. As evidenced by simulations of the baseline extreme wet year scenario, releases early in the year are mostly dictated by the flood-control curve, leaving reduced options later in the year if climate or hydrology indicates a warmer or dryer fall than expected. The increase in the November 1 cold water pool for the all-out-the-uppermost TCD schedule when comparing a wet year with a dry fall to an extreme wet year occurs because the lower streamflow inputs later in the year for the wet year with a dry fall mean that total outflow later in the year (especially during the Fall X2 operations) does not need to be as high, which reduces the withdrawal of water above the temperature thresholds.

Air temperature increases from projected future climate scenarios for the simulated extreme dry year worsened degree-day exceedances and reduced November 1 cold pool volumes for all operations schedules, but only worsened degree-day exceedances in the wet year for releases out the uppermost gates. These simulations only increased air temperatures and used the same hydrology as the baseline extreme wet and dry simulations. Thus, climate change effects could be greater because climate change studies in the region have indicated that snowpack and streamflow timing and quantities are likely to change (Cayan et al. 2001; Stewart et al. 2004). Trinh et al. (2016) recently developed the watershed environmental hydrology hydro-climate model (WEHY-HCM) for the Upper Sacramento River Basin, which could be used to generate inflows to Shasta Reservoir under different climate change scenarios, although that model does not appear to simulate water temperatures as well. The results of the study presented here indicate that climate change scenarios make management of downstream temperatures more challenging in all aspects, exacerbating the difficulties of both maintaining cold water reserve and limiting

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degree-day exceedances. Additional changes due to altered snowpack and hydrology would likely compound the inability of TCD operations to control downstream temperatures for salmon.

Buccola et al. (2016) used *W*2 with a hydrology model driven by high-emissions climate data from three global circulation models to examine in-lake and downstream water temperature impacts for Detroit Lake in Oregon. The hydrology model simulated a 23% reduction in annual average discharge and 1°C increase in mean annual stream temperatures for inflow tributaries to the reservoir with climate change inputs. *W*2 simulations indicated that annual average in-lake and outflow water temperatures were 1.1 and 1.5°C higher, respectively with climate change. Similar to the study presented here, Buccola et al. (2016) found that altered reservoir operations resulted in outlet water temperatures closer to desired temperature targets, especially when hypothetical structures to alter release elevations were modeled. Previous *W*2 simulations by Hanna et al. (1999) and Bartholow et al. (2001) for Shasta Reservoir also found that managing release elevations with the TCD resulted in fewer degree-day exceedances than using historical operations through bypass and penstock intake elevations.

Simulation results are for releases from one elevation at a time except when transitioning between elevations (Fig. 4). The project team did additional simulations with blended releases from multiple elevations at the same time, but results indicated minimal improvement with blended releases for either reducing degree-day exceedances or increasing cold pool storage. Thus, simulation results indicate that operational constraints due to the multiple reservoir objectives are most limiting for meeting downstream temperature requirements. These results agree with previous simulations to evaluate the effect of TCD operations at Shasta Lake by Bartholow et al. (2001) and Saito et al. (2001) that indicated that climate and hydrology had stronger relationships to downstream temperatures and in-reservoir ecosystem dynamics than reservoir operations.

Some assumptions were made when developing this model and analysis. One assumption is that the outflows through the TCD will follow a flow-weighted average through all open TCD gates. This was a necessary assumption for a 2D model where it is not possible to separate flows out through different gates across the face of the dam at the same elevation. In addition, gate configurations were simulated to only change once a day. In reality, reservoir managers use an iterative process when deciding which gates to operate at a given time (R. Yaworsky, personal communication, 2014), and leakage does occur through closed gates (Higgs and Vermeyen 1999).

Another assumption involved the calculation of the cold water pool in the analysis presented here. The volume of the cold water pool was calculated based on the water temperatures below 9°C for the segment at the face of the dam and it was assumed that layers below 9°C at the dam were this temperature throughout the reservoir. Calculations of the volumes of cells below 9°C indicated that differences in volume between volume based on temperature distributions at the dam versus throughout the reservoir were minimal. The cold pool calculation was done weekly due to the time required to estimate the cold pool volume over time. The 2D representation of Shasta Lake in *W*2 would allow calculation of the cold pool reserve by model segment, but due to the coarseness of model layers in the deeper parts of the reservoir the weekly calculations by layer at the dam seemed a reasonable calculation. The lowest vertical layers of the reservoir were originally developed at 6-m intervals because cold water reserve calculation was not the original intent of the model when it was developed to simulate reservoir hydrodynamics and estimate outflow temperatures. Thus, further use of the model for understanding options for managing cold pool volume should consider increasing vertical layer resolution.

Future work involves linking *W*2 with a riverine temperature model (Pike et al. 2013) to enable translation of water temperatures from Shasta Lake further downstream through the Sacramento River. It will also allow more realistic evaluation of meeting temperature criteria that are not actually set directly below the dam as was evaluated for the project described here. *W*2 can also be linked with ecological models as was done by Saito et al. (2001) to understand the effects of reservoir operations on fisheries throughout the Sacramento River system.

Conclusions

The combined use of stochastic generation of extreme hydroclimate conditions with 2D reservoir modeling of Shasta Lake indicates that reservoir managers have limited options in extremely wet and extremely dry years to manage reservoir temperature releases for downstream fisheries. Productive interaction between modelers and reservoir managers helped to design simulations that were informative to decision makers regarding reservoir operations in extreme hydroclimate conditions. Trade-offs between meeting desired temperatures downstream and preserving the reservoir's cold pool for releases for endangered winter-run Chinook salmon were evident in scenario runs. Expected climate change will likely exacerbate these challenges.

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